

Disproof of conjecture 00000000125: the prime-trace count in $\mathrm{SL}_2(\mathbb{F}_p)$ is not $\Theta(p^2/\log p)$

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1 The conjecture and its reading

Statement (English, verbatim). Conjecture: The number of elements of $\mathrm{SL}_2(\mathbb{F}_p)$ whose trace has absolute value a prime is asymptotic to $c \cdot p^2/\log p$, with c explicit from a Sato–Tate type density for the trace distribution.

The Chinese text states the same claim, with the same normalisation $c \cdot p^2/\log p$ (it spells the name of the density “Sato–Tarte”).

Reading. The trace of $A \in \mathrm{SL}_2(\mathbb{F}_p)$ is an element of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. For “its absolute value is a prime” to make sense it must be lifted to an integer. We use an arbitrary lift L and then the two standard lifts:

- the *centered* lift: the representative in $(-p/2, p/2]$ (Lean: `ZMod.valMinAbs`);
- the *standard* lift: the representative in $[0, p)$ (Lean: `ZMod.val`).

Definition 1. For a lift L and a prime p , let $N_L(p) = \#\{A \in \mathrm{SL}_2(\mathbb{F}_p) : |L(\mathrm{tr} A)| \text{ is a prime number}\}$, where $\mathrm{tr} A = A_{11} + A_{22}$.

The claim “ $N(p) \sim c p^2/\log p$ ” is read as $p \rightarrow \infty$ through the primes, with $\log$ the natural logarithm. We refute it in three forms, each for both standard lifts and for every real c (positive or not), which covers the hedge “with c explicit”:

1. the ratio $N(p)/(c p^2/\log p)$ does not tend to 1;
2. $N(p)$ is not asymptotically equivalent to $c p^2/\log p$ (Mathlib’s `IsEquivalent`);
3. $N(p)$ is not even $O(p^2/\log p)$, so $N(p) = \Theta(p^2/\log p)$ also fails.

Conventions. In Lean, “ $p \rightarrow \infty$ through the primes” is the filter `atTop` on the subtype $\{p : \mathbb{N} \mid p \text{ prime}\}$, ordered as a subset of $\mathbb{N}$. Lean division satisfies $x/0 = 0$. This matters only for $c = 0$, where the ratio is identically 0 and cannot tend to 1; the proof does not use it. $N_L(p)$ is defined with `Nat.card` of a subtype of Mathlib’s `Matrix.SpecialLinearGroup (Fin 2) (ZMod p)`, which is a finite group for prime p . The trace is Mathlib’s `Matrix.trace`. A non-negative integer n is prime in the usual sense (`Nat.Prime`); $|\cdot|$ is `Int.natAbs`.

2 Proof

Lemma 2 (trace-2 family). *Let R be a commutative ring, $a \in R$ and $b \in R^\times$. The matrix*

$$T(a, b) = \begin{pmatrix} a & b \\ -(a-1)^2 b^{-1} & 2-a \end{pmatrix}$$

has determinant 1 and trace 2, and $(a, b) \mapsto T(a, b)$ is injective.

Proof. $\det T(a, b) = a(2-a) + (a-1)^2 b b^{-1} = 2a - a^2 + a^2 - 2a + 1 = 1$, and $\text{tr } T(a, b) = a + (2-a) = 2$. The entries in positions (1, 1) and (1, 2) recover a and b . $\square$

Lemma 3 (lower bound). *Let p be prime, and suppose $|L(2)|$ is prime. Then $N_L(p) \geq p(p-1)$.*

Proof. Every $T(a, b)$ with $a \in \mathbb{F}_p$, $b \in \mathbb{F}_p^\times$ lies in $\text{SL}_2(\mathbb{F}_p)$ and has trace 2, so it is counted by $N_L(p)$. There are $p(p-1)$ such pairs, and the map is injective. $\square$

Lemma 4 (lifts of 2). *For $p \geq 4$ the centered lift of $2 \in \mathbb{F}_p$ is 2; for $p \geq 3$ the standard lift of 2 is 2. In particular, $|L(2)| = 2$ is prime for both lifts and every prime $p \geq 5$.*

Proof. $2 \leq p/2$ when $p \geq 4$, and $2 < p$ when $p \geq 3$. $\square$

Theorem 5. *Let L be a lift such that $|L(2)|$ is prime for all large primes p (for example, either standard lift). Then $N_L(p) \neq O(p^2/\log p)$ as $p \rightarrow \infty$ through the primes. Hence, for every $c \in \mathbb{R}$, $N_L(p) \not\sim c p^2/\log p$ and $N_L(p)/(c p^2/\log p) \not\rightarrow 1$.*

Proof. Suppose $N_L(p) \leq C p^2/\log p$ for all large primes p . Take a prime $p \geq 3$ that is large enough for this bound, for $\log p > 2C$, and for $|L(2)|$ to be prime. Write $x = p$ and $\ell = \log x > 0$. By Lemma 2, $x(x-1) \leq N_L(p) \leq C x^2/\ell$, so $\ell x(x-1) \leq C x^2 < \ell x^2/2$. Then $x-1 < x/2$, that is, $x < 2$, which is a contradiction.

If $N_L \sim c p^2/\log p$, then $N_L = O(c p^2/\log p) = O(p^2/\log p)$, which we just excluded. If the ratio tends to 1, then $N_L \sim c p^2/\log p$ (Mathlib's `isEquivalent_of_tendsto_one`, which needs no side condition). $\square$

Remark 6 (the true order; no Lean). *For p odd and $t \in \mathbb{F}_p$, the number of $A \in \text{SL}_2(\mathbb{F}_p)$ with $\text{tr } A = t$ is $p^2 + \chi(t^2 - 4)p$, where χ is the Legendre symbol with $\chi(0) = 0$. To see this, write $A = \begin{pmatrix} a & b \\ c & t-a \end{pmatrix}$, so $bc = a(t-a) - 1$. A nonzero value of bc has $p-1$ solutions (b, c) , the value 0 has $2p-1$, and exactly $1 + \chi(t^2 - 4)$ values of a make $a(t-a) = 1$. We checked this numerically for primes up to 53. Under the centered lift the qualifying traces are $\pm q$ for primes $q \leq (p-1)/2$; under the standard lift they are the primes $q < p$. So $N(p) = p^2 \#\{\text{qualifying } t\} + O(p^2/\log p)$, which is of order $p^3/\log p$ by the prime number theorem. The trace distribution on $\text{SL}_2(\mathbb{F}_p)$ is uniform up to a relative error $O(1/p)$, so a “Sato–Tate type density” does not enter the leading constant. We do not formalize this remark; the disproof uses only Lemma 2.*

3 Lean formalization

Lean 4 (toolchain v4.33.1) with Mathlib. The file is `lean/Conjecture125/Basic.lean` (190 lines), namespace `C125`. It contains no `sorry`, no `native_decide` and no added axioms.

- `SL2 p := Matrix.SpecialLinearGroup (Fin 2) (ZMod p); tr A := Matrix.trace` of the underlying matrix.

- `centeredLift n x := x.valMinAbs; standardLift n x := (x.val : Int).`
- `N L p := Nat.card {A : SL2 p // (L p (tr A)).natAbs.Prime}.`
- `PrimeNat := {p : Nat // p.Prime},` with filter `atTop`.
- `Conjecture125 L := exists c : Real, 0 < c /\ Tendsto (fun p : PrimeNat => (N L p : Real) / (c * p^2 / Real.log p)) atTop (nhds 1).`
- `traceTwo, tr_traceTwo, traceTwo_injective:` Lemma 1.
- `lower_bound:` Lemma 2, $p(p-1) \leq N_L(p)$.
- `centeredLift_two, standardLift_two:` Lemma 3.
- `not_isBig0, not_isEquivalent` (every real c), and `not_tendsto_ratio` (no real c): the theorem, for a general lift L .
- **Main theorem** `conjecture125_false:` $\sim \text{Conjecture125 centeredLift} \wedge \sim \text{Conjecture125 standardLift}$ together with the failure of $N = O(p^2/\log p)$ for both lifts.

Axiom audit. `#print axioms C125.conjecture125_false` (and likewise for `not_isEquivalent` and `not_tendsto_ratio`) prints `[proptext, Classical.choice, Quot.sound]`.
`lake build` succeeds.